# Fusang: Tardigrade Edition

Alignment-free phylogenetic inference with k-mer frequency vector cosine distances

**INPUT**

**Unaligned FASTA (DNA or protein)**

*no alignment step*

→

**K-MER VECTORS**

**Spaced + contiguous k-mer frequency vectors**

*auto k,gap selection*
*multi-k ensemble: no k choice*

→

**COSINE DISTANCE**

**Pairwise cosine distance matrix**

*boundary classifier: homogeneous vs structured*

→

**TREE**

**NJ / FastME**
**BIONJ+BNNI → Newick**

*simplified ≤500 taxa*
*DCM >500 taxa*

## Validated across simulation and real data

### Indel-rich accuracy

n=200, indel rate 0.02, vs true tree (nRF ↓)

marginally better than FastTree2 (p=0.0507, borderline n.s.)

**5% more accurate than IQ-TREE2 GTR ($p=9.21\times10^{-5}$)**

### Real-data validation

AFproject community benchmark, ETE3 nRF ↓

**Fish mitogenomes (n=25, ~17 kb)** **0.045**

community-benchmark ceiling — ties MSA+ML

**13 mtDNA genes (168–1,866 bp)** **0.451**

Fusang–FastTree2 agreement (13 loci)

**SwissTree proteins (11 families)** **0.329**

k-mer cosine vs Co-phylog 0.727 (p=0.0010)

**no k selection at any scale: 500 bp genes → 17 kb genomes**

### Fast, scalable, ready to use

**10,000 taxa** **69.8 s**

single CPU core

**1,000 taxa** **5.2 s**

simplified pipeline

**boundary classifier** **150/150**

E2E scenario routing

**pre-compiled Windows/Linux binaries · MIT license**

**github.com/zhanglknt/fusang-tardigrade**